

Step 1.7 Implementation Report

- Healthcare AI QA Engineering - CareFlow Health
- Document Type: Step Implementation Report
- Status: PASS
- Scope: Acceptance Criteria -> Test Scenario only

Objective

- Define canonical business-level test scenarios from approved acceptance criteria, with complete traceability and prioritization for future manual, BDD, and UI automation execution.

Architecture and Context

- Project remains frontend-only. No backend, API, API automation, RAG, MCP, LangGraph, or Playwright/Cucumber code was implemented in this step.
- All references and scenario data are synthetic. No PHI or real healthcare integrations were introduced.

Requirements-to-Test Lifecycle

- BRD -> PRD -> Epic -> User Story -> Acceptance Criteria -> Test Scenario
- Traceability document was extended to include Acceptance Criteria-to-Test Scenario mapping and coverage summary.

Design Approach

- Used business-focused scenario design with stable IDs (TS-<EPIC>-<SEQUENCE>).
- Applied happy path + negative/validation + state/accessibility/responsive pattern per user story where appropriate.
- Maintained explicit metadata for future BDD/Playwright conversion.

Scenario Categories

- Positive, Negative, Validation, Boundary, Error Handling, Role/Authorization UI, Accessibility, Responsive, Cross-Browser, Regression, and additional UI state/navigation categories where applicable.

Prioritization Model

- Regression priorities assigned using business criticality, user impact, risk, frequency, and regression importance.
- Distribution: P0=16, P1=40, P2=40, P3=0.

Smoke vs Regression

- Smoke scenarios were intentionally restricted to critical journeys.
- Smoke scenarios: 6.
- All scenarios include regression priority.

BDD and Playwright Relationship

- Each scenario includes BDD intent in Given/When/Then style.
- Each scenario includes automation notes for future Playwright implementation without writing selectors or code in this step.

Coverage Results

- Total User Stories: 32
- Total Acceptance Criteria: 109
- Total Test Scenarios: 96
- Acceptance Criteria covered: 109
- Acceptance Criteria without scenario coverage: 0
- Orphan scenarios: 0
- Positive: 32, Negative: 32, Validation: 32, Boundary: 12, Error Handling: 32
- Accessibility: 32, Responsive: 32, Cross-Browser: 32

Validation Results

- Source documents reviewed completely.
- Unique scenario IDs verified.
- Traceability completeness and gap checks completed.
- No out-of-scope implementation artifacts introduced.

Interview Q&A Themes

- Scenario vs test case
- Scenario vs acceptance criterion
- Coverage decision model
- Smoke/regression strategy
- RAG/MCP/LangGraph readiness
- Frontend-only QA implications

Lessons Learned

- Structured metadata at scenario stage significantly improves downstream automation and AI reasoning readiness.
- Coverage quality depends on risk-based grouping, not mechanical one-criterion-to-one-scenario expansion.

Future Steps

- Next recommended step: Step 1.8 - Test Case Definition.